

HUMAN CONNECTOME · SESSION ANALYSIS REPORT

Session ID: session_1781286508279_p40vc
Period: 2026-06-12T16:14:09.489Z → 2026-06-12T17:48:28.202Z
Wall-clock duration: 5658.7s | Ticks: 5,096 | Estimated tick interval: ~1110.4ms
High-confidence thoughts (≥75%): 0

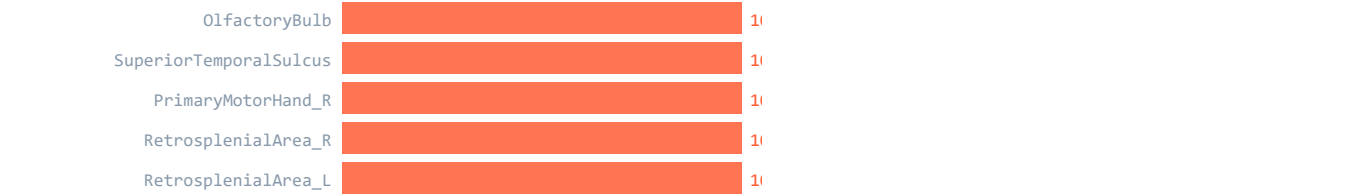
Scientific Validity Notice: Claim language used throughout: "emergent behavior in a brain-inspired embodied simulation." Shannon entropy is normalized as $H/\log(N_{\text{regions}})$; max=1.0. Session duration uses wall-clock time. Pearson r is correlation only – NOT transfer entropy. $r=\pm 1.000$ flagged as possible computation artifact. STDP = STDP-inspired population plasticity (Wilson-Cowan rate model). Platform biologically constrained by HCP-MMP1.0, Allen Brain Atlas, and Brainnetome Atlas data.

⚠ Saturation Warning: 5 region(s) averaged >90% activation: OlfactoryBulb, SuperiorTemporalSulcus, PrimaryMotorHand_R, RetrosplenialArea_R, RetrosplenialArea_L. Verify homeostatic scaling before publishing emergence claims.

SESSION METRICS

DURATION 5658.7 seconds (wall-clock)	TICKS 5,096 simulation steps	SHANNON ENTROPY 0.848 $H/\log(N)$ · max=1.0	PLASTICITY INDEX 0.147 STDP-inspired · pop. rate
---	---	--	---

NEURAL ACTIVITY · TOP 15 REGIONS BY AVG ACTIVATION



#	REGION	AVG ACTIVATION	SATURATION FLAG
1	OlfactoryBulb	100%	⚠ SATURATED – verify
2	SuperiorTemporalSulcus	100%	⚠ SATURATED – verify
3	PrimaryMotorHand_R	100%	⚠ SATURATED – verify
4	RetrosplenialArea_R	100%	⚠ SATURATED – verify
5	RetrosplenialArea_L	100%	⚠ SATURATED – verify

STDP-INSPIRED POPULATION PLASTICITY · WEIGHT CHANGES

Wilson-Cowan rate model. Eligibility-trace approximation. NOT literal spike-timing plasticity.



CONNECTION	DELTA	TYPE
PrefrontalCortex->Thalamus	-0.7000	LTD
PrefrontalCortex->BasalGanglia	-0.7000	LTD
PrefrontalCortex->AnteriorCingulateCortex	-0.7000	LTD
PrefrontalCortex->DorsalACC	-0.7000	LTD
SupplementaryMotorArea->MotorCortex	-0.7000	LTD

CONNECTION	DELTA	TYPE
Amygdala->Thalamus	-0.7000	LTD
Amygdala->Insula	-0.7000	LTD
Brainstem->Thalamus	-0.7000	LTD
Brainstem->Cerebellum	-0.7000	LTD
DeepCerebellarNuclei->Thalamus	-0.7000	LTD

EMERGENT THOUGHT LOG (≥75% CONFIDENCE MULTI-REGION CO-ACTIVATION)

Thoughts require ≥3 regions simultaneously co-active above threshold. Silence periods are valid scientific data.

CORRELATION ANALYSIS (PEARSON R · NOT TRANSFER ENTROPY)

Pearson r (lag-1) = symmetric linear association. Transfer entropy = directional information-theoretic measure. These are NOT interchangeable.

REGION PAIR	PEARSON R	VALIDITY
PrefrontalCortex->Hippocampus	0.0263	OK
PrefrontalCortex->Thalamus	0.0000	OK
PrefrontalCortex->Amygdala	0.0000	OK
PrefrontalCortex->BasalGanglia	0.0000	OK
PrefrontalCortex->AnteriorCingulateCortex	0.0000	OK

EXECUTIVE SUMMARY

- [1] Arousal Pattern: The avatar maintained a low average arousal of 31% across 5090 ticks (peak: 37%). Low arousal suggests dominance of inhibitory tone, likely reflecting resting-state default mode network activity.
- [2] Emotional Trajectory: The avatar's affective state was relatively stable over the session. Valence stability suggests balanced excitatory/inhibitory regulation, a hallmark of healthy homeostatic control.
- [3] Synaptic Plasticity (STDP): The strongest synaptic potentiation occurred at VentralTegmentalArea->NucleusAccumbens, NucleusAccumbens->VentralPallidum, OlfactoryBulb->Hippocampus. These connections strengthened because both pre- and post-synaptic regions fired in correlated temporal proximity – the core mechanism of Hebbian learning. This implies the avatar's neural architecture is self-organizing around the most frequently co-activated circuits.
- [4] Behavioral Patterns: The dominant behavioral state was "RESTING" (resting: 5016, fearful: 81, motivated: 79). Resting state predominance suggests the simulation spent significant time in low-drive default mode, common during low-complexity or low-arousal conditions.
- [5] Cardiac/ANS Summary: Average heart rate was 60 BPM. Low resting heart rate reflects strong parasympathetic tone with dominant hypothalamic and serotonergic modulation – the autonomic baseline of a calm, regulated state.

BEHAVIORAL EVENTS

TICK	TYPE	DESCRIPTION
T5095	DROP	NAC silencing -> reward expectancy drop
T5095	DROP	Hippocampal quiescence -> consolidation phase
T5095	CASCADE	STRATEGY_SHIFT: repeated failure on 'pause' (3x) -> depressed fronto-striatal pathway -> promoting 'investigate'
T5094	DROP	PontineTegmentum drop
T5094	SURGE	NAC dopamine surge -> reward prediction signal
T5094	DROP	PFC deactivation -> reduced cognitive control
T5093	DROP	VTA suppression -> reward signal attenuated
T5093	SURGE	Hippocampal theta burst -> memory encoding/retrieval
T5093	SURGE	PFC activation -> executive override of limbic response

TICK	TYPE	DESCRIPTION
T5092	SURGE	VTA activation -> mesolimbic dopamine release
T5091	DROP	Parietal silencing -> spatial awareness reduced
T5091	DROP	Visual cortex suppression -> perceptual gating
T5091	DROP	Insula silencing -> visceral awareness reduced
T5091	DROP	PFC deactivation -> reduced cognitive control
T5090	SURGE	Ventral pallidum activation -> reward integration output
T5090	DROP	dACC quiescence -> reduced cognitive conflict monitoring
T5090	SURGE	Hippocampal theta burst -> memory encoding/retrieval
T5090	SURGE	PFC activation -> executive override of limbic response
T5089	SURGE	Dorsal ACC surge -> cognitive control, conflict signaling
T5089	CASCADE	STRATEGY_SHIFT: repeated failure on 'pause' (3x) -> depressed fronto-striatal pathway -> promoting 'approach'